1. **Symmetry Rule**

$$\binom{n}{k} = \binom{n}{n-k}$$

2. **Factoring In**

$$\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1}$$

3. **Sum Over k**

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

4. **Sum Over n**

$$\sum_{m=0}^{n} \binom{m}{k} = \binom{n+1}{k+1}$$

5. **Sum Over n and k**

$$\sum_{k=0}^{m} \binom{n+k}{k} = \binom{n+m+1}{m}$$

6. **Sum of the Squares**

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2 = \binom{2n}{n}$$

7. **Weighted Sum**

$$1 \cdot \binom{n}{1} + 2 \cdot \binom{n}{2} + \cdots + n \cdot \binom{n}{n} = n \cdot 2^{n-1}$$

8. **Connection with Fibonacci Numbers**

$$\binom{n}{0} + \binom{n-1}{1} + \cdots + \binom{n-k}{k} + \cdots + \binom{0}{n} = F_{n+1}$$